Sheet 318/333 Brighton and Worthing 1: 50 000 Scale Geology Series

Answers

What date was this latest version published?

2006

When was the original geological survey published?

1864

One of the original surveyors was W B Dawkins – look him up – what is his connection with Manchester?

William Boyd Dawkins became Curator of the Manchester Museum and he was the first Professor of Geology at Owens College which went on to become the University of Manchester. We have part of his original book collection in our Study. When it opens again you are welcome to view these historic books.

What scale is this map?

1;50,000

What distance would 2cm on the map represent on the ground?

1 cm = 50,000 cm = 500m so 2cm = 500X2 = 1000m or 1km

Describe what is meant by Bedrock and Superficial Deposits?

Bedrock are the lithified (rocky) rocks which are grouped into Formations. They are shown in the Generalized Vertical Sections in stratigraphic order with the oldest at the bottom of the section.

Superficial Deposits are the unconsolidated (loose) deposits and they are grouped according to their type e.g. beach and tidal flat deposits. They are not shown in the key in the order of superposition (age order)

In which quadrant (N, S, E, W) of the map do the oldest rock formations outcrop?

The oldest formations are the Wadhurst Clay Formation – you can find this in the NE section of the map at the northern end of Section 2 (GR3271 2220) The Wadhurst Clay Formation belongs within the Wealden Group and is Lower Cretaceous in age.

Do you think the beds in the formations are in horizontal layers or they dipping i.e. inclined from their original horizontal attitude? Explain your answer and show what evidence you used to arrive at your answer.

The beds are gently dipping (less than 8 degrees) towards the South.

Evidence: Form the dip arrows on the map; by looking at the vertical cross sections and by the fact that the maps shows younger formations outcropping towards the South in the direction of dip.

Find the Gault Formation. What range of heights above sea level is the land surface where the Gault outcrops?

The Gault Formation forms a clay “vale” or flat lying area. Heights 53 and 34m.